

5 **Draft Standard for**
6 **Local and metropolitan area networks—**

7 **Link Aggregation**

8 **Amendment 1:**
9 **YANG for Link Aggregation**

10 Prepared by the

11 **Time-Sensitive Networking (TSN) Task Group of IEEE 802.1**

12 Sponsor

13 **LAN/MAN Standards Committee**

14 **of the**

15 **IEEE Computer Society**

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² The text of this draft, following these cover pages, is that of P802.1AXdz as of the final Working Group
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⁴ **YANG modules**

⁵ The YANG modules specified by this standard are attached to the draft pdf as plain text (UTF-8) .yang files.

⁶

⁷

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2 **Abstract:** This amendment to IEEE Std 802.1AX-2020 specifies a Unified Modeling Language
3 (UML)-based model and YANG modules for Link Aggregation configuration and status reporting.

4 **Keywords:** Aggregated Link, Aggregator, Distributed Resilient Network Interconnect, DRNI,
5 interconnect, Link Aggregation, Link Aggregation Group, local area network, management,
6 Network-Network Interface, NNI, YANG.

7

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1 Introduction

This introduction is not part of IEEE Std 802.1AXdz™-20XX, IEEE Standard for Local and metropolitan area networks—Link Aggregation—Amendment 1: YANG for Link Aggregation.

2 IEEE Std 802.1AXdz™-202X: YANG for Link Aggregation specifies a Unified Modeling Language
3 (UML)-based model and YANG modules for Link Aggregation configuration and status reporting

4 This standard contains state-of-the-art material. The area covered by this standard is undergoing evolution.
5 Revisions are anticipated within the next few years to clarify existing material, to correct possible errors, and
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¹ 2. Normative references

² *Insert the following references into Clause 2 in alphanumeric order:*

³ IETF RFC 6241, Network Configuration Protocol (NETCONF), June 2011.

⁴ IETF RFC 6991, Common YANG Data Types, July 2013

⁵ IETF RFC 7950, The YANG 1.1 Data Modeling Language, August 2016.

⁶ IETF RFC 8343, A YANG Data Model for Interface Management, March 2018

⁷

¹ 4. Acronyms and abbreviations

² *Insert the following acronyms into Clause 4 in alphabetical order:*

³ NETCONF Network Configuration Protocol

⁴ UML® Unified Modeling Language™¹

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1 5. Conformance

2 *Insert the following text (items j and k) after item i) in the lettered list of 5.3.2*

3 5.3.2 Link Aggregation options

- 4 j) Support YANG modules for the management of Link Aggregation capabilities (Clause 10).
- 5 k) Support YANG modules for the management of DRNI capabilities (Clause 10).

1 *Insert Clause 10 after Clause 9 as follows:*

2 **10. YANG Data Model**

3 This clause specifies the YANG (IETF RFC 7950) data model that provides control and status monitoring of
4 systems and system components that implement functionality specified in this standard.

5 This clause:

- 6 a) Introduces the YANG framework that governs the naming and hierarchy of configuration and
7 operational data structures in the data models, and the modeling of network interfaces (10.1).
- 8 b) Describes the information data model and its relationship to the operational processes and managed
9 objects specified in the other clauses of this standard, and provides a representation, similar to the
10 Unified Modeling Language (UML), of relevant objects (10.2).
- 11 c) Describes the structure of the YANG data model, which comprises three YANG modules (10.3).
- 12 d) Includes a relationship description of other modules imported in YANG modules (10.4)
- 13 e) Reviews security considerations applicable to each of the modules, with specific reference to data
14 nodes in the YANG modules that compose the YANG data model (10.5).
- 15 f) Includes each of the YANG modules and its data schema (10.6).

16 **10.1 YANG Framework**

17 The YANG framework applies hierarchy in the following areas:

- 18 a) The uniform resource name (URN), as specified in 802d.
- 19 b) The YANG objects form a hierarchy of configuration and operational data structures that specify the
20 YANG model.

21 **10.2 Information Model for Link Aggregation Management**

22 The YANG objects are based on the managed objects in Clause 7. A UML-like representation of the
23 management model is provided in the following subclauses.

24 NOTE—OMG® UML2.5² [B11] conventions together with C++ language constructs are used in this clause as a
25 representation to convey model structure and relationships..

26 The purpose of an UML-like³ diagram is to express the model design on a single piece of paper. The
27 structure of the UML-like representation shows the name of the object followed by a list of properties for the
28 object. The properties indicate its type and accessibility. The UML-like representation is meant to express
29 simplified semantics for the properties. It is not meant to provide the specific datatype as used to encode the
30 object in either MIB or YANG. In the UML-like representation, a box with a white background represents
31 information that comes from sources outside of the IEEE. A box with a gray background represents objects
32 that are specified by this IEEE Standard.

33 The YANG hierarchical structure that incorporates the Link Aggregation YANG modules supported by this
34 standard is represented by Figure 10-1. In the figures in this clause, items that are shaded gray are described
35 in this document, items with no background shading are specified elsewhere. The YANG data model is

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³A description of the UML-like diagrams used in this clause is provided at <https://1.ieee802.org/uml-like-diagrams>.

1 realized in three YANG modules. One module *ieee802-dot1ax-types* provides data types that are needed by
2 the Link Aggregation configuration and monitoring objects. The *ieee802-dot1ax-linkagg* module provides
3 the Link Aggregation, Aggregation Port, and Aggregator configuration and monitoring objects. The
4 *ieee802-dot1ax-drni* module provides the Distributed Resilient Network Interface (DRNI) configuration and
5 monitoring objects. The Link Aggregation and DRNI capabilities are not only applicable to IEEE Std
6 802.1Q bridges, but also (for example) end stations, and routers.

7

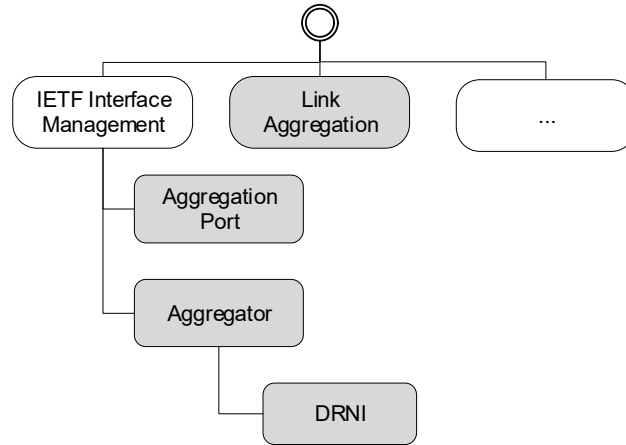


Figure 10-1—YANG root hierarchy with Link Aggregation YANG modules

8 10.2.1 Link Aggregation model

9 The Link Aggregation Configuration and Monitoring Objects in Figure 10-2 show the objects that are
10 applicable on a system supporting link aggregation, and the interfaces supporting link aggregation.

11 The objects on the system consist of a list of aggregation systems, and a list of key groups. A key group
12 is the set of aggregators and aggregation ports that share the same system priority, system mac address, and
13 aggregation key, and therefore can potentially form a Link Aggregation Group. Each key group has a unique
14 combination of actor-admin-key and actor-system, and includes parameters that have the same value for any
15 aggregator and/or aggregation port that have the same actor-admin-key and actor-system.

16 NOTE—See 6.3.5 and 6.3.6 for a description of the identification of Aggregation Ports that can potentially form a Link
17 Aggregation Group, and Figure 6-15 for examples.

18 Each Aggregation Port is an interface (e.g. of type ethernetCsmacd) augmented with an “aggport” presence
19 container as shown on the right side of Figure 10-2. Each Aggregator is an interface to the entire Link
20 Aggregation Group (using type *ieee8023adlag*⁴) augmented with a “lag” container as shown on the left side
21 of Figure 10-2. The binding of Aggregation Ports to Aggregators is a dynamic function of the Link
22 Aggregation Control Protocol. The Aggregation Ports attached to an Aggregator can be read from the *lower-*
23 *layer-if* attribute of the Aggregator interface. The Aggregator to which an Aggregation Port is attached can
24 be read through the *higher-layer-if* attribute of the Aggregation Port interface. An Aggregation Port will only
25 transmit and respond to LACPDUs when the interface “enabled” leaf is true, and will only attach to an
26 Aggregator when the Aggregator “enabled” leaf is true. The interface “oper-status” leaf of an Aggregation
27 Port will be “up” when the Aggregation Port is attached to an Aggregator and the “collecting” and
28 “distributing” bits of the “actor-oper-state” leaf are both true. The interface “oper-status” leaf on an
29 Aggregator will be “up” when it has at least one attached Aggregation Port that is “up”.

⁴Link Aggregation was originally published as IEEE Std 802.3ad-2000. It was moved to the 802.1 Working Group in 2008 and subsequent revisions have been published as IEEE Std 802.1AX.

1

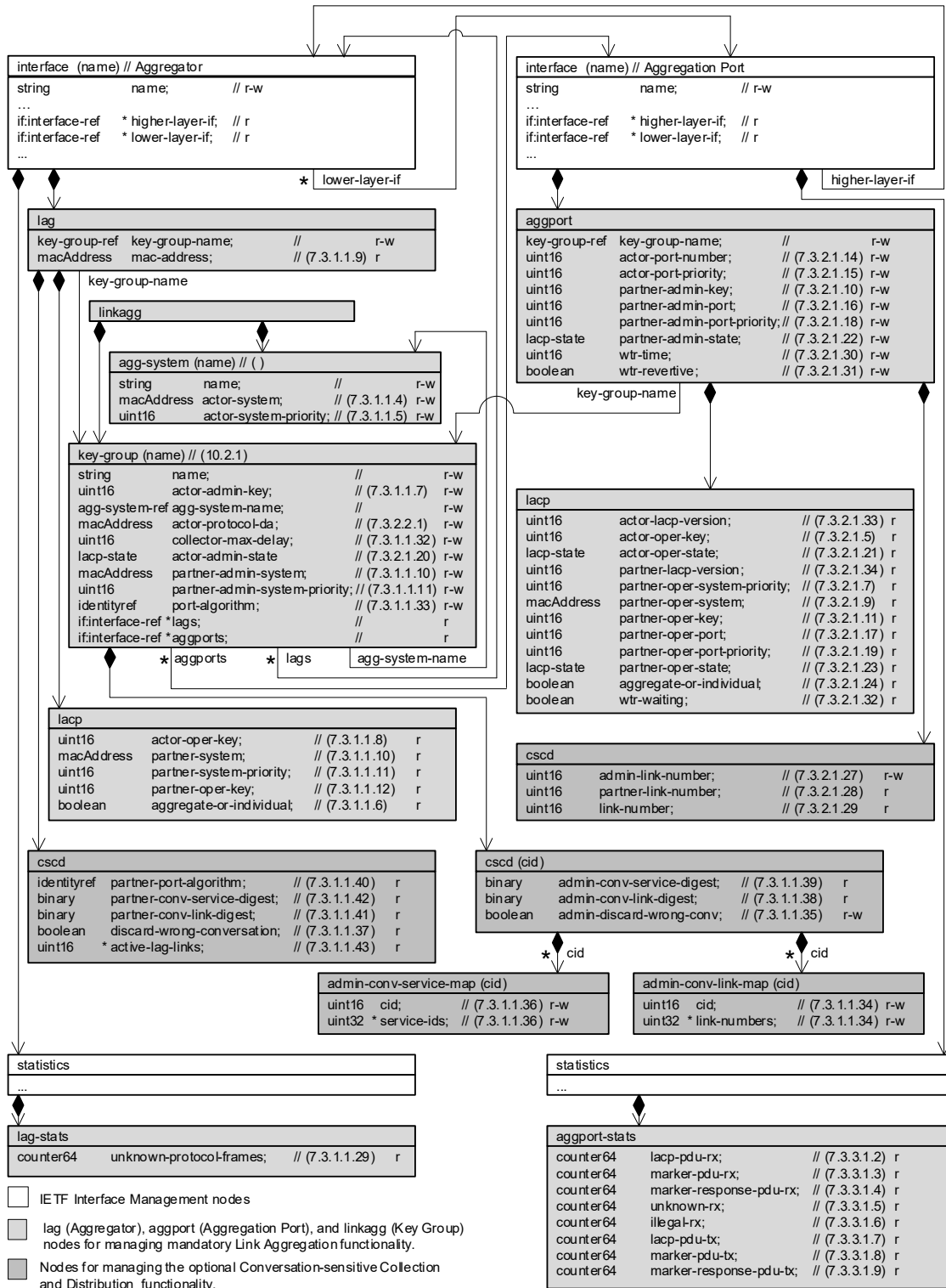


Figure 10-2—Link Aggregation Configuration and Monitoring Objects

1 10.2.2 DRNI model

2 The DRNI Configuration and Monitoring Objects in Figure 10-3 show the objects that are applicable to an
3 Aggregator augmented with a “drni” container.

4

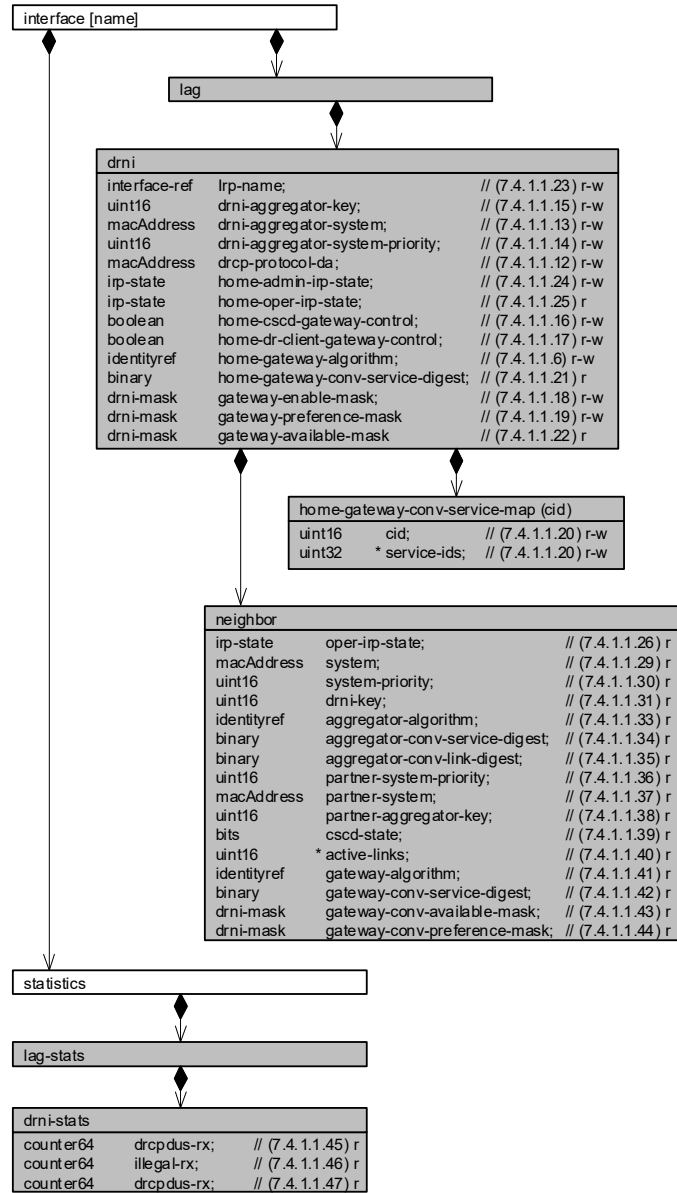


Figure 10-3—DRNI Configuration and Monitoring Objects

10.3 Structure of the Link Aggregation YANG Model

The YANG data model specified in this standard is comprised of three YANG modules. A summary of the modules contained in this clause is represented in Table 10-1.

Table 10-1—Structure of the YANG modules

Module	Subclause	Notes
ieee802-dot1ax-types	10.7.1	Type definitions used for Link Aggregation YANG.
ieee802-dot1ax-linkagg	10.7.2	Link Aggregation Management
ieee802-dot1ax-drni	10.7.3	DRNI Management

In the YANG modules below, if any discrepancy between the DESCRIPTION text and the corresponding specification in any other part of this standard occurs, the specifications outside this clause take precedence.

10.4 Relationship to other YANG modules

This clause describes how the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules are related to the YANG modules that are imported.

10.4.1 IEEE 802.1AX Types Module

The *ieee802-dot1ax-types* module provides reusable types that are used by the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* modules.

10.4.2 IETF YANG Types Module

The *ietf-yang-types* YANG module (IETF RFC 6991) contains a set of derived YANG types. This document leverages counter64.

10.4.3 IETF Interfaces YANG Module

The *ietf-interfaces* YANG module (IETF RFC 8343) contains a set of YANG definitions for managing network interfaces. This document augments an *ietf-interfaces:interface* with aggregation-port or aggregator data nodes..

10.4.4 IEEE 802 Types Module

The *ieee802-types* module provides reusable types that are used in IEEE 802 standards.

NOTE—The type for mac-addresses defined in *ieee802-types* has a pattern that allows upper and lower case letters. To avoid issues with string comparison, it is suggested to only use upper case for the letters in the hexadecimal numbers. Implementers using code comparing MAC addresses are advised that there is still an issue with a difference between the IETF mac-address definition and the IEEE mac-address definition.

10.5 Security Considerations

The YANG modules specified in this clause are designed to be accessed via a network configuration protocol (e.g., NETCONF protocol). In the case of NETCONF, the lowest NETCONF layer is the secure transport layer and the mandatory to implement secure transport is SSH. The NETCONF access control model provides the means to restrict access for particular NETCONF users to a pre-configured subset of all available NETCONF protocol operations and content.

It is the responsibility of a system's implementor and administrator to ensure that the protocol entities in the system that support NETCONF, and any other remote configuration protocols that make use of these YANG modules, are properly configured to allow access only to those users who have legitimate rights to read or write data nodes. This standard does not specify how the credentials of those users are to be stored or validated.

10.5.1 Security considerations of the *ieee802-dot1ax* YANG modules

There are several management objects specified in the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules that are configurable (i.e., read-write) and/or operational (i.e., read-only). Such objects could be considered sensitive or vulnerable in some network environments. A network configuration protocol, such as NETCONF (IETF RFC 6241), can support protocol operations that can edit or delete YANG module configuration data (e.g., edit-config, delete-config, copy-config). If this is done in a non-secure environment without proper protection, then negative effects on the network operation are possible.

The following containers, and the objects in these containers, of the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules can be manipulated to interfere with the operation of the Link Aggregation Control Protocol (LACP) or the Distributed Relay Control Protocol (DRCP). This could, for example, cause network instability and result in the loss of service for a large number of end users.

- `dot1ax-linkagg/linkagg/key-group`
- `dot1ax-linkagg/linkagg/key-group/cscd`
- `dot1ax-linkagg/linkagg/key-group/cscd/admin-conv-service-map`
- `dot1ax-linkagg/linkagg/key-group/cscd/admin-conv-link-map`
- `dot1ax-linkagg/aggport`
- `dot1ax-linkagg/aggport/cscd/admin-link-number`
- `dot1ax-linkagg/lag`
- `dot1ax-drni/lag/drni`
- `dot1ax-drni/lag/drni/home-gateway-conv-service-map`

Some of the readable data in this YANG module could be considered sensitive or vulnerable in some network environments. It is important to control all types of access (e.g., including NETCONF get, get-config operations) to these objects and possibly to even encrypt the values of these objects when sending them over the network. For example, the system name and other information about the remote systems could provide information about the configuration and topology of the network and could be considered a privacy threat.

10.6 YANG schema tree definitions^{5,6}

A simplified graphical representation of the data model is used in this document. The meaning of the symbols in these diagrams is as follows:

- Brackets "[" and "]" enclose list keys.
- Abbreviations before data node names: "rw" means configuration (read-write), and "ro" means state data (read-only).
- Symbols after data node names: "?" means an optional node, "!" means a presence container, and "*" denotes a list and leaf-list.
- Parentheses enclose choice and case nodes, and case nodes are also marked with a colon (":").
- Ellipsis ("...") stand for contents of subtrees that are not shown.

10.6.1 Schema for the *ieee802-dot1ax-linkagg* YANG module

```

module: ieee802-dot1ax-linkagg
  +--rw linkagg
    +--rw agg-system* [name]
      | +--rw name string
      | +--rw actor-system? ieee:mac-address
      | +--rw actor-system-priority? uint16
    +--rw key-group* [name]
      +--rw name string
      +--rw actor-admin-key uint16
      +--rw agg-system-name agg-system-ref
      +--rw actor-protocol-da? ax:bridge-reserved-addresses
      +--rw collector-max-delay? uint16
      +--rw actor-admin-state? ax:lacp-state
      +--rw partner-admin-system? ieee:mac-address
      +--rw partner-admin-system-priority? uint16
      +--rw port-algorithm? identityref
      +--ro lags* if:interface-ref
      +--ro aggports* if:interface-ref
      +--rw cscd {ax:cscd}?
        +--rw admin-conv-service-map {ax:sid-map}?
          | +--rw (method)?
          |   +--:(pattern)
          |   | +--rw pattern? identityref
          |   +--:(cid-list)
          |   +--rw cid-list* [cid]
          |     +--rw cid uint16
          |     +--rw service-ids* uint32
          +--ro admin-conv-service-digest? binary
          +--rw admin-conv-link-map
            +--rw (method)?
            | +--:(pattern)
            | | +--rw pattern? identityref
            | +--:(cid-list)
            | +--rw cid-list* [cid]
            |   +--rw cid uint16
            |   +--rw link-numbers* uint16
            +--ro admin-conv-link-digest? binary
          +--rw admin-discard-wrong-conv? enumeration {ax:dwc}?
    augment /if:interfaces/if:interface:
      +--rw lag
        +--rw key-group-name key-group-ref
        +--ro mac-address? ieee:mac-address
        +--ro lacp
          | +--ro actor-oper-key? uint16

```

⁵Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

⁶An ASCII version of the YANG module(s) can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.


```

1      | +--ro partner-system?          ieee:mac-address
2      | +--ro partner-system-priority? uint16
3      | +--ro partner-oper-key?       uint16
4      | +--ro aggregate-or-individual? boolean
5      +--ro cscd {ax:cscd}?
6          +--ro partner-port-algorithm? identityref
7          +--ro partner-conv-service-digest? binary
8          +--ro partner-conv-link-digest? binary
9          +--ro discard-wrong-conversation? boolean
10         +--ro active-lag-links*      uint16
11 augment /if:interfaces/if:interface/if:statistics:
12     +--ro lag-stats
13         +--ro unknown-protocol-frames? yang:counter64
14 augment /if:interfaces/if:interface:
15     +--rw aggport!
16         +--rw key-group-name          key-group-ref
17         +--rw actor-port-number?      uint16
18         +--rw actor-port-priority?    uint16
19         +--rw partner-admin-key?      uint16
20         +--rw partner-admin-port?     uint16
21         +--rw partner-admin-port-priority? uint16
22         +--rw partner-admin-state?    ax:lacp-state
23         +--rw wtr-time?               uint16
24         +--rw wtr-revertive?          boolean
25         +--ro lacp
26             | +--ro actor-lacp-version? uint8
27             | +--ro actor-oper-key?     uint16
28             | +--ro actor-oper-state?   ax:lacp-state
29             | +--ro partner-lacp-version? uint8
30             | +--ro partner-oper-system-priority? uint16
31             | +--ro partner-oper-system? ieee:mac-address
32             | +--ro partner-oper-key?   uint16
33             | +--ro partner-oper-port?  uint16
34             | +--ro partner-oper-port-priority? uint16
35             | +--ro partner-oper-state? ax:lacp-state
36             | +--ro aggregate-or-individual? boolean
37             | +--ro wtr-waiting?        boolean
38         +--rw cscd {ax:cscd}?
39             +--rw admin-link-number?    uint16
40             +--ro partner-link-number?  uint16
41             +--ro link-number?          uint16
42 augment /if:interfaces/if:interface/if:statistics:
43     +--ro aggport-stats
44         +--ro lacp-pdu-rx?              yang:counter64
45         +--ro marker-pdu-rx?            yang:counter64
46         +--ro marker-response-pdu-rx?   yang:counter64
47         +--ro unknown-rx?               yang:counter64
48         +--ro illegal-rx?               yang:counter64
49         +--ro lacp-pdu-tx?              yang:counter64
50         +--ro marker-pdu-tx?            yang:counter64
51         +--ro marker-response-pdu-tx?   yang:counter64
52

```

53 10.6.2 Schema for the *ieee802-dot1ax-drni* YANG module

```

54 module: ieee802-dot1ax-drni
55
56 augment /if:interfaces/if:interface/dot1ax:lag:
57     +--rw drni!
58         +--rw irp-name                  if:interface-ref
59         +--rw drni-aggregator-key?      uint16
60         +--rw drni-aggregator-system?    ieee:mac-address
61         +--rw drni-aggregator-system-priority? uint16
62         +--rw drcp-protocol-da?          ax:bridge-reserved-addresses
63         +--rw home-admin-irp-state?      irp-state
64         +--ro home-oper-irp-state?       irp-state
65         +--rw home-cscd-gateway-control? boolean
66         +--rw home-dr-client-gateway-control? boolean
67         +--rw home-gateway-algorithm?    identityref
68         +--rw home-gateway-conv-service-map {ax:sid-map}?
69         | +--rw (method)?

```

```

1      |      +---:(pattern)
2      |      | +--rw pattern?      identityref
3      |      +---:(cid-list)
4      |      | +--rw cid-list* [cid]
5      |      | +--rw cid          uint16
6      |      | +--rw service-ids* uint32
7      +---ro home-gateway-conv-service-digest?  binary
8      +---rw gateway-enable-mask
9      | +--rw cid-str*      string
10     | +--rw invert-mask?  boolean
11     +---rw gateway-preference-mask
12     | +--rw cid-str*      string
13     | +--rw invert-mask?  boolean
14     +---ro gateway-available-mask
15     | +---ro cid-str*      string
16     | +---ro invert-mask?  boolean
17     +---ro neighbor
18         +---ro oper-irp-state?      irp-state
19         +---ro system?              ieee:mac-address
20         +---ro system-priority?     uint16
21         +---ro drni-key?            uint16
22         +---ro aggregator-algorithm? identityref
23         +---ro aggregator-conv-service-digest?  binary
24         +---ro aggregator-conv-link-digest?    binary
25         +---ro partner-system-priority?        uint16
26         +---ro partner-system?                ieee:mac-address
27         +---ro partner-aggregator-key?        uint16
28         +---ro cscd-state?                    bits
29         +---ro active-links*                  uint16
30         +---ro gateway-algorithm?             identityref
31         +---ro gateway-conv-service-digest?   binary
32         +---ro gateway-available-mask
33         | +---ro cid-str*      string
34         | +---ro invert-mask?  boolean
35         +---ro gateway-preference-mask
36         | +---ro cid-str*      string
37         | +---ro invert-mask?  boolean
38     augment /if:interfaces/if:interface/if:statistics/dot1ax:lag-stats:
39         +---ro drni-stats
40         | +---ro drcpdus-rx?  yang:counter64
41         | +---ro illegal-rx?  yang:counter64
42         | +---ro drcpdus-tx?  yang:counter64
43

```

44 10.7 YANG modules

45 10.7.1 The *ieee802-dot1ax-types* YANG module

```

46 module ieee802-dot1ax-types {
47   yang-version 1.1;
48   namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-types";
49   prefix dot1ax-types;
50
51   organization
52     "IEEE 802.1 Working Group";
53   contact
54     "WG-URL: http://www.ieee802.org/1/
55      WG-Email: stds-802-1-1@ieee.org
56
57     Contact: IEEE 802.1 Working Group Chair
58     Postal: C/O IEEE 802.1 Working Group
59            IEEE Standards Association
60            45 Hoes Lane
61            Piscataway, NJ 08854
62            USA
63     E-mail: stds-802-1-chairs@ieee.org";
64   description
65     "Common types used within 802.1AX Link Aggregation modules.
66

```

```
1      Copyright (C) IEEE (2025).
2
3      This version of this YANG module is part of IEEE Std 802.1AX;
4      see the standard itself for full legal notices.";
5
6      revision 2025-03-11 {
7          description
8              "Published as part of IEEE Std 802.1AXdz-202y.
9              The following reference statement identifies each referenced IEEE
10             Standard as updated by applicable amendments.";
11          reference
12              "IEEE Std 802.1AX-2020 Link Aggregation:
13              IEEE Std 802.1AXdz-202y.";
14      }
15
16      feature csdc {
17          description
18              "Conversation Sensitive Collection and Distribution (CSCD)
19              is supported.";
20          reference
21              "5.3.2, 6.6 of IEEE Std 802.1AX";
22      }
23
24      feature dwc {
25          description
26              "The Discard Wrong Conversation option in CSCD is
27              supported.";
28          reference
29              "5.3.2, 6.6 of IEEE Std 802.1AX";
30      }
31
32      feature sid-map {
33          description
34              "The Service ID to Conversation ID map is supported for
35              use by distribution algorithms.";
36      }
37
38      feature c-vid-dist-alg {
39          description
40              "Supports distribution based on C-VIDs.";
41      }
42
43      feature s-vid-dist-alg {
44          description
45              "Supports distribution based on S-VIDs.";
46      }
47
48      feature i-sid-dist-alg {
49          description
50              "Supports distribution based on I-SIDs.";
51      }
52
53      feature te-sid-dist-alg {
54          description
55              "Supports distribution based on TE-SIDs.";
56      }
57
58      feature flow-hash-dist-alg {
59          description
60              "Supports distribution based on Flow Hash.";
61      }
62
63      feature basic-link-maps {
64          description
65              "Pre-defined admin-conv-link-map configurations to distribute
66              packets on either one or two active links, with any
67              remaining links available as standby.";
68      }
69
70      identity distribution-algorithm {
71          description
72              "Each distribution algorithm is identified by a sequence of
```

```
1      4 octets, structured as shown in Figure 8-1. Distribution
2      algorithm identifiers are used by network administrators to
3      select between algorithms and, in Conversation-sensitive
4      LACP and (if supported) Distributed Resilient Network
5      Interconnect (DRNI) operation, to check whether partners
6      and neighbors are using the same algorithm.
7
8      This identity is intended to serve as base identity, not
9      to be directly referenced.
10
11     Vendor specific, combination (ex: multi-layer), and other
12     customized distribution algorithms can be created as
13     their own identities in their own YANG files, derived from
14     this imported base type. When the identity description
15     specifies a 4 octet value, this value is used for cscd
16     purposes, and included in LACPDUs. Otherwise the identity
17     is used purely for local selection of a distribution
18     algorithm, and the unspecified distribution algorithm is
19     used for cscd purposes.";
20     reference
21       "8.1, 8.2 of IEEE Std 802.1AX";
22   }
23
24   identity unspecified {
25     base distribution-algorithm;
26     description
27       "The 'Unspecified distribution algorithm' identifier has
28       been reserved for use when the algorithm is unknown (or
29       is not advertised).
30       The algorithm identifier is 00-80-C2-00.";
31     reference
32       "Table 8-1 of IEEE Std 802.1AX";
33   }
34
35   identity c-vids-nomap {
36     if-feature "c-vid-dist-alg";
37     base distribution-algorithm;
38     description
39       "Distribution based on C-VIDs (8.2.1). No Service ID
40       mapping table is used.
41       The algorithm identifier is 00-80-C2-01.";
42     reference
43       "Table 8-1 of IEEE Std 802.1AX";
44   }
45
46   identity c-vids-map {
47     if-feature "c-vid-dist-alg and sid-map";
48     base distribution-algorithm;
49     description
50       "Distribution based on C-VIDs (8.2.1). A Service ID
51       mapping table is used.
52       The algorithm identifier is 00-80-C2-81.";
53     reference
54       "Table 8-1 of IEEE Std 802.1AX";
55   }
56
57   identity s-vids-nomap {
58     if-feature "s-vid-dist-alg";
59     base distribution-algorithm;
60     description
61       "Distribution based on S-VIDs (8.2.2). No Service ID
62       mapping table is used.
63       The algorithm identifier is 00-80-C2-02.";
64     reference
65       "Table 8-1 of IEEE Std 802.1AX";
66   }
67
68   identity s-vids-map {
69     if-feature "s-vid-dist-alg and sid-map";
70     base distribution-algorithm;
71     description
72       "Distribution based on S-VIDs (8.2.2). A Service ID
```

```
1      mapping table is used.
2      The algorithm identifier is 00-80-C2-82.";
3  reference
4      "Table 8-1 of IEEE Std 802.1AX";
5  }
6
7  identity i-sids-nomap {
8      if-feature "i-sid-dist-alg";
9      base distribution-algorithm;
10     description
11         "Distribution based on I-SIDs (8.2.3). No Service ID
12         mapping table is used.
13         The algorithm identifier is 00-80-C2-03.";
14     reference
15         "Table 8-1 of IEEE Std 802.1AX";
16 }
17
18 identity i-sids-map {
19     if-feature "i-sid-dist-alg and sid-map";
20     base distribution-algorithm;
21     description
22         "Distribution based on I-SIDs (8.2.3). A Service ID
23         mapping table is used.
24         The algorithm identifier is 00-80-C2-83.";
25     reference
26         "Table 8-1 of IEEE Std 802.1AX";
27 }
28
29 identity te-sids-nomap {
30     if-feature "te-sid-dist-alg";
31     base distribution-algorithm;
32     description
33         "Distribution based on TE-SIDs (8.2.4). No Service ID
34         mapping table is used.
35         The algorithm identifier is 00-80-C2-04.";
36     reference
37         "Table 8-1 of IEEE Std 802.1AX";
38 }
39
40 identity te-sids-map {
41     if-feature "te-sid-dist-alg and sid-map";
42     base distribution-algorithm;
43     description
44         "Distribution based on TE-SIDs (8.2.4). A Service ID
45         mapping table is used.
46         The algorithm identifier is 00-80-C2-84.";
47     reference
48         "Table 8-1 of IEEE Std 802.1AX";
49 }
50
51 identity flow-hash-nomap {
52     if-feature "flow-hash-dist-alg";
53     base distribution-algorithm;
54     description
55         "Distribution based on Flow Hash (8.2.5). No Service ID
56         mapping table is used.
57         The algorithm identifier is 00-80-C2-05.";
58     reference
59         "Table 8-1 of IEEE Std 802.1AX";
60 }
61
62 identity flow-hash-map {
63     if-feature "flow-hash-dist-alg and sid-map";
64     base distribution-algorithm;
65     description
66         "Distribution based on Flow Hash (8.2.5). A Service ID
67         mapping table is used.
68         The algorithm identifier is 00-80-C2-85.";
69     reference
70         "Table 8-1 of IEEE Std 802.1AX";
71 }
72
```

```
1 identity link-map-patterns {
2   description
3     "Base identity for patterns filling the admin-conv-link-map.
4     This identity is intended to serve as base identity, not
5     to be directly referenced. Use of the identity allows vendors
6     to augment the module with vendor specified patterns,";
7   reference
8     "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
9 }
10
11 identity one-plus-n {
12   if-feature "basic-link-maps";
13   base link-map-patterns;
14   description
15     "Provides active/standby behavior with one active link and
16     up to 63 standby links. All packets are mapped to whichever
17     link that is active in the aggregation has the lowest link
18     number. If that link fails, all packets are mapped to the
19     lowest link number of the remaining links that are active
20     in the aggregation.
21     The map consists of an identical entry for each CID, with a
22     list of numbers from 1 to 64.
23     Supports any number of links in the aggregation (up to 64)
24     with link numbers between 1 and 64. Any links in the
25     aggregation with a link number greater than 64 are not
26     used.";
27 }
28
29 identity two-plus-n {
30   if-feature "basic-link-maps";
31   base link-map-patterns;
32   description
33     "Provides basic load-balancing on two active links with up
34     to 62 standby links. All packets with an even CID value are
35     mapped to the link with the lowest link number in the
36     aggregation, and all packets with an odd CID value are
37     mapped to the link with the highest link number in the
38     aggregation.
39     The map consists of an identical entry for each even CID
40     containing a list of numbers from 1 to 64, and an identical
41     entry for each odd CID containing a list of numbers from 64
42     to 1.
43     Supports any number of links in the aggregation (up to 64)
44     with link numbers between 1 and 64. Any links in the
45     aggregation with a link number greater than 64 are not
46     used.";
47 }
48
49 identity service-map-patterns {
50   description
51     "Base identity for patterns filling the admin-conv-service-map.
52     This identity is intended to serve as base identity, not
53     to be directly referenced. Use of the identity allows vendors
54     to augment the module with vendor specified patterns,";
55   reference
56     "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
57 }
58
59 typedef lacp-state {
60   type bits {
61     bit lacp-activity {
62       position 1;
63       description
64         "Provides administrative control over when LACPDU's are
65         transmitted. A value of '1' indicates Active mode where
66         LACPDU's are sent regardless of partner's lacp-activity
67         value. A value of '0' indicates Passive mode where
68         LACPDU's are sent only when the partner's lacp-activity
69         value is '1' (partner is in Active mode).";
70     }
71     bit lacp-timeout {
72       position 2;
```

```

1      description
2          "Provides administrative control over the frequency of
3          received LACPDUs. A value of '1' indicates Short Timeout
4          (so partner uses frequent transmission). A value of '0'
5          indicates Long Timeout (so partner can use infrequent
6          transmission).";
7      }
8      bit aggregation {
9          position 3;
10         description
11             "Provides administrative control over whether this
12             Aggregation Port can be in a LAG with more than one
13             member. A value of '1' indicates the port can be
14             aggregated with other ports. A value of '0' indicates
15             the port can only be a solitary link.";
16     }
17     bit synchronization {
18         position 4;
19         description
20             "The Synchronization state of the MUX state machine.";
21     }
22     bit collecting {
23         position 5;
24         description
25             "The Collecting state of the MUX state machine.";
26     }
27     bit distributing {
28         position 6;
29         description
30             "The Distributing state of the MUX state machine.";
31     }
32     bit defaulted {
33         position 7;
34         description
35             "Indicates the port is using the partner-admin values
36             to select an Aggregator.";
37     }
38     bit expired {
39         position 8;
40         description
41             "The Expired state of the Receive state machine.";
42     }
43 }
44 description
45     "LACP state values as transmitted in LACPDUs.
46     Administrative control over the values of lacp-activity,
47     lacp-timeout, aggregation, and (in partner-admin-lacp-state)
48     synchronization is allowed.
49     The remaining bits are read-only.";
50 reference
51     "6.4.1, 6.4.2.3 of IEEE Std 802.1AX";
52 }
53
54 typedef bridge-reserved-addresses {
55     type enumeration {
56         enum nearest-customer-bridge {
57             description
58                 "Reserved MAC address 01-C2-80-00-00";
59         }
60         enum slow-protocols-multicast {
61             description
62                 "Reserved MAC address 01-C2-80-00-02";
63         }
64         enum nearest-non-tpmr-bridge {
65             description
66                 "Reserved MAC address 01-C2-80-00-03";
67         }
68         enum nearest-bridge {
69             description
70                 "Reserved MAC address 01-C2-80-00-0E";
71         }
72     }
73 }

```

```
1  description
2    "Group addresses that are filtered by some or all types of
3    bridges in order to limit the scope of propagation of PDUs
4    in a Bridged Network.";
5  reference
6    "8.6.3 of IEEE Std 802.1Q";
7  }
8
9  grouping link-map {
10   description
11     "Specifies the contents of the admin-conv-link-map.";
12   reference
13     "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
14   choice method {
15     default "cid-list";
16     description
17       "Provides two ways to specify the map contents.";
18     leaf pattern {
19       type identityref {
20         base link-map-patterns;
21       }
22       description
23         "Use a predefined pattern to fill the map.";
24     }
25     list cid-list {
26       key "cid";
27       description
28         "Data structure to map a Conversation Identifier
29         (CID) to a Link Number. Each entry consists of a CID
30         and a list of link numbers that can potentially be
31         selected for that CID. An empty list of link-numbers
32         means that no links are selected for the CID.";
33       leaf cid {
34         type uint16 {
35           range "0..4095";
36         }
37         description
38           "Port Conversation Identifier";
39       }
40       leaf-list link-numbers {
41         type uint16;
42         description
43           "List of zero or more Link Numbers that can
44           potentially be selected for distribution of frames
45           with this CID.";
46       }
47     }
48   }
49 }
50
51 grouping service-map {
52   description
53     "Specifies the contents of the admin-conv-service-map.";
54   reference
55     "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
56   choice method {
57     default "cid-list";
58     description
59       "Provides two ways to specify the map contents.";
60     leaf pattern {
61       type identityref {
62         base service-map-patterns;
63       }
64       description
65         "Use a predefined pattern to fill the map.";
66     }
67     list cid-list {
68       key "cid";
69       description
70         "Data structure to map service identifiers to
71         conversation identifiers. Each entry consists of a
72         Conversation ID (CID) and a list of zero or more
```



```
1      Service Identifiers (SIDs) that map to it. An empty
2      list of SIDs means there are no SIDs that map to this
3      CID, and results in the same behavior as not having an
4      entry for this CID.";
5  leaf cid {
6      type uint16 {
7          range "0..4095";
8      }
9      description
10         "Port Conversation Identifier";
11  }
12  leaf-list service-ids {
13      type uint32;
14      description
15         "List of zero or more SIDs that map to the CID.";
16  }
17  }
18  }
19  }
20 }
21
22
```

23 10.7.2 The *ieee802-dot1ax-linkagg* YANG module

```
24 module ieee802-dot1ax-linkagg {
25     yang-version 1.1;
26     namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-linkagg";
27     prefix dot1ax;
28
29     import ieee802-dot1ax-types {
30         prefix ax;
31     }
32     import ieee802-types {
33         prefix ieee;
34     }
35     import ietf-yang-types {
36         prefix yang;
37     }
38     import ietf-interfaces {
39         prefix if;
40     }
41     import iana-if-type {
42         prefix ianaif;
43     }
44
45     organization
46         "IEEE 802.1 Working Group";
47     contact
48         "WG-URL: http://www.ieee802.org/1/
49         WG-Email: stds-802-1-l@ieee.org
50
51         Contact: IEEE 802.1 Working Group Chair
52         Postal: C/O IEEE 802.1 Working Group
53                 IEEE Standards Association
54                 45 Hoes Lane
55                 Piscataway, NJ 08854
56                 USA
57         E-mail: stds-802-1-chairs@ieee.org";
58     description
59         "This YANG module describes the configuration model for Link
60         Aggregation, as specified in IEEE Std 802.1AX, including Link
61         Aggregation Control Protocol (LACP) and Conversation Sensitive
62         Collection and Distribution.
63
64         Copyright (C) IEEE (2025).
65
66         This version of this YANG module is part of IEEE Std 802.1AX;
67         see the standard itself for full legal notices.";
68
69     revision 2025-03-11 {
```

```
1   description
2     "Published as part of IEEE Std 802.1AXdz-202y.
3     The following reference statement identifies each referenced IEEE
4     Standard as updated by applicable amendments.";
5   reference
6     "IEEE Std 802.1AX-2020 Link Aggregation:
7     IEEE Std 802.1AXdz-202y.";
8 }
9
10 typedef key-group-ref {
11   type leafref {
12     path "/dotlax:linkagg/dotlax:key-group/dotlax:name";
13   }
14   description
15     "This type is used by aggregators and aggregation ports to
16     reference an entry in the key-group list.";
17 }
18
19 typedef agg-system-ref {
20   type leafref {
21     path "/dotlax:linkagg/dotlax:agg-system/dotlax:name";
22   }
23   description
24     "This type is used in the key-group list entries to
25     reference an entry in the agg-system list.";
26 }
27
28 augment "/if:interfaces/if:interface" {
29   when "derived-from-or-self(if:type,'ianaif:ieee8023adLag')" {
30     description
31       "Applies to interfaces representing a LAG.";
32   }
33   description
34     "Augment Interface with Aggregator parameters.";
35   container lag {
36     description
37       "Contains the configuration and status information that
38       allows an instance of an Aggregator to be managed.";
39     leaf key-group-name {
40       type key-group-ref;
41       mandatory true;
42       description
43         "Specifies the entry in the link-aggregation key-group
44         list to which this aggregator is assigned.";
45     }
46     leaf mac-address {
47       type ieee:mac-address;
48       config false;
49       description
50         "The MAC address assigned to the Aggregator.";
51       reference
52         "7.3.1.1.9 of IEEE Std 802.1AX";
53     }
54     container lacp {
55       config false;
56       description
57         "Contains aggregator LACP operational data.";
58       leaf actor-oper-key {
59         type uint16;
60         description
61           "The current operational value of the Key for the
62           Aggregator. The meaning of particular Key
63           values is of local significance.";
64         reference
65           "7.3.1.1.8 of IEEE Std 802.1AX";
66       }
67       leaf partner-system {
68         type ieee:mac-address;
69         description
70           "The value of the MAC address portion of the
71           System Identifier for the current
72           protocol partner of this Aggregator. A value
```

```
1         of zero indicates that there is no known partner.
2         If the aggregation is manually configured, this
3         System identifier value is assigned by the
4         local System.";
5     reference
6         "7.3.1.1.10 of IEEE Std 802.1AX";
7 }
8 leaf partner-system-priority {
9     type uint16;
10    description
11        "Indicates the priority value portion of the current
12        Partner's System Identifier. If the aggregation is
13        manually configured, this System Priority value is
14        assigned by the local System.";
15    reference
16        "7.3.1.1.11 of IEEE Std 802.1AX";
17 }
18 leaf partner-oper-key {
19     type uint16;
20    description
21        "The current operational value of the Key for the
22        Aggregators current protocol Partner. If the
23        aggregation is manually configured, this Key value
24        is assigned by the local System.";
25    reference
26        "7.3.1.1.12 of IEEE Std 802.1AX";
27 }
28 leaf aggregate-or-individual {
29     type boolean;
30    description
31        "Indicates whether the Aggregator represents an
32        Aggregate (TRUE) or an Individual link (FALSE).";
33    reference
34        "7.3.1.1.6 of IEEE Std 802.1AX";
35 }
36 }
37 container cscd {
38     if-feature "ax:cscd";
39     config false;
40     description
41         "Aggregator parameters obtained by the operation of LACP
42         supporting CSCD.";
43     leaf partner-port-algorithm {
44         type identityref {
45             base ax:distribution-algorithm;
46         }
47         description
48             "Operational value of the distribution algorithm in
49             use by the LACP Partner.";
50         reference
51             "7.3.1.1.40 of IEEE Std 802.1AX";
52     }
53     leaf partner-conv-service-digest {
54         type binary;
55         description
56             "The MD5 Digest of the admin-conv-service-map in use
57             by the LACP Partner.";
58         reference
59             "7.3.1.1.42, 6.6.3.1 of IEEE Std 802.1AX";
60     }
61     leaf partner-conv-link-digest {
62         type binary;
63         description
64             "The MD5 Digest of the admin-conv-link-map in use
65             by the LACP Partner.";
66         reference
67             "7.3.1.1.41, 6.6.3.1 of IEEE Std 802.1AX";
68     }
69     leaf discard-wrong-conversation {
70         type boolean;
71         description
72             "The operational value that determines whether an
```

```

1         Aggregator discards a frame that is collected
2         from an Aggregation Port that is different from the
3         Aggregation Port to which the Aggregator would
4         distribute a frame with the same Port Conversation
5         ID.";
6     reference
7         "7.3.1.1.37, 6.6 of IEEE Std 802.1AX";
8 }
9 leaf-list active-lag-links {
10     type uint16;
11     description
12         "A list, possibly empty, of the operational
13         link-number of each Aggregation Port active
14         (i.e. Collecting) on this Aggregator.";
15     reference
16         "7.3.1.1.43 of IEEE Std 802.1AX";
17 }
18 }
19 }
20 }
21
22 augment "/if:interfaces/if:interface/if:statistics" {
23     when '../dot1ax:lag' {
24         description
25             "Applies to aggregators.";
26     }
27     description
28         "Augment interface statistics with aggregator statistics.";
29     container lag-stats {
30         config false;
31         description
32             "Contains the set of stats associated with the
33             Aggregator.";
34         leaf unknown-protocol-frames {
35             type yang:counter64;
36             description
37                 "A count of data frames discarded on reception by all
38                 ports that are (or have been) members of the
39                 aggregation, due to the detection of an unknown Slow
40                 Protocols PDU (7.3.3.1.5)";
41             reference
42                 "7.3.1.1.29 of IEEE Std 802.1AX";
43         }
44     }
45 }
46
47 augment "/if:interfaces/if:interface" {
48     description
49         "Augment interface model with Aggregation port
50         configuration nodes.";
51     container aggport {
52         presence
53             "When present, this interface supports Link Aggregation";
54         description
55             "Contains Aggregation Port configuration related nodes,
56             which provides the basic management controls necessary
57             to allow an instance of an Aggregation Port to be managed,
58             for the purposes of Link Aggregation.";
59         leaf key-group-name {
60             type key-group-ref;
61             mandatory true;
62             description
63                 "Specifies the entry in the link-aggregation key-group
64                 list to which this aggregation-port is assigned.";
65         }
66         leaf actor-port-number {
67             type uint16 {
68                 range "1..65535";
69             }
70             description
71                 "The port number assigned to the Aggregation Port.
72                 The port number is communicated in LACPDUs as the

```

```
1      Actor Port.
2      This leaf provides the ability to administratively
3      override the initial value provided by the system.";
4      reference
5      "7.3.2.1.14, 6.4.6 of IEEE Std 802.1AX";
6  }
7  leaf actor-port-priority {
8      type uint16;
9      default "0x8000";
10     description
11     "The priority value assigned to this Aggregation Port.";
12     reference
13     "7.3.2.1.15, 6.4.6 of IEEE Std 802.1AX";
14 }
15 leaf partner-admin-key {
16     type uint16 {
17         range "1..65535";
18     }
19     description
20     "The current administrative value of the Key for the
21     protocol Partner. The assigned value is used, along
22     with the value of port-partner-admin-system-priority,
23     partner-admin-system, partner-admin-port, and
24     partner-admin-port-priority, in order to achieve
25     manually configured aggregation.
26     The default value is the initial value of actor-port-
27     number provided by the system.";
28     reference
29     "7.3.2.1.10 of IEEE Std 802.1AX";
30 }
31 leaf partner-admin-port {
32     type uint16 {
33         range "1..65535";
34     }
35     description
36     "The current administrative value of the port number for
37     the protocol Partner. The assigned value is used, along
38     with the value of partner-admin-system-priority,
39     partner-admin-system, port-partner-admin-key, and
40     partner-admin-port-priority, in order to achieve
41     manually configured aggregation.
42     The default value is the initial value of actor-port-
43     number provided by the system.";
44     reference
45     "7.3.2.1.16 of IEEE Std 802.1AX";
46 }
47 leaf partner-admin-port-priority {
48     type uint16;
49     default "0";
50     description
51     "The current administrative value of the port priority
52     for the protocol Partner. The assigned value is used,
53     along with the value of partner-admin-system-priority,
54     partner-admin-system, partner-admin-key, and
55     partner-admin-port, in order to achieve manually
56     configured aggregation.";
57     reference
58     "7.3.2.1.18 of IEEE Std 802.1AX";
59 }
60 leaf partner-admin-state {
61     type ax:lacp-state {
62         bit lacp-activity {
63             position 1;
64             description
65             "When the LACP_Activity state is set to '0', the
66             transmission of LACPDUs is controlled by the
67             actor-admin-state.LACP_Activity.";
68         }
69         bit lacp-timeout {
70             position 2;
71             description
72             "When the LACP_Timeout is set to '0', LACPDUs
```

```

1         are transmitted at the Slow_Periodic_Time.";
2     }
3     bit aggregation {
4         position 3;
5         description
6             "When the Aggregation state is set to '0', this
7             port cannot be aggregated with any other port.";
8     }
9     bit synchronization {
10        position 4;
11        description
12            "When the Synchronization state is set to '0', this
13            port cannot become active.";
14    }
15 }
16 default "synchronization";
17 description
18     "Provides administrative control over the partner's
19     LACP_Activity, LACP_Timeout, Aggregation, and
20     Synchronization state when the partner's information is
21     unknown (i.e. no LACPDUs are received from the partner).";
22 reference
23     "7.3.2.1.22, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
24 }
25 leaf wtr-time {
26     type uint16 {
27         range "0..32767";
28     }
29     default "0";
30     description
31         "The wait-to-restore (WTR) period, in seconds, that
32         needs to elapse between an Aggregation Port on a LAG
33         coming up (Port_Operational becoming TRUE) and being
34         permitted to become active (transmitting and
35         receiving frames) on the LAG.";
36     reference
37         "7.3.2.1.30 of IEEE Std 802.1AX";
38 }
39 leaf wtr-revertive {
40     type boolean;
41     default "true";
42     description
43         "Controls revertive or non-revertive mode of operation.
44         When TRUE, the Aggregation Port can become active as
45         soon as the wait-to-restore timer expires regardless of
46         the state of other links in the LAG.
47         When FALSE, the Aggregation Port cannot become active
48         unless there are no other links that can become active
49         in the LAG. The default value is TRUE.";
50     reference
51         "7.3.2.1.31 of IEEE Std 802.1AX";
52 }
53 container lacp {
54     config false;
55     description
56         "Contains Aggregation port LACP operational related
57         nodes.";
58     leaf actor-lacp-version {
59         type uint8;
60         description
61             "The version number transmitted in LACPDUs on this
62             Aggregation Port";
63         reference
64             "7.3.2.1.33 of IEEE Std 802.1AX";
65     }
66     leaf actor-oper-key {
67         type uint16;
68         description
69             "The current operational value of the Key for the
70             Aggregation Port. The meaning of particular Key values
71             is of local significance.";
72         reference

```

```
1      "7.3.2.1.5 of IEEE Std 802.1AX";
2  }
3  leaf actor-oper-state {
4      type ax:lacp-state;
5      description
6          "The operational value of the Actor_State as
7           transmitted in LACPDUs.";
8      reference
9          "7.3.2.1.21, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
10 }
11 leaf partner-lacp-version {
12     type uint8;
13     description
14         "The version number in the LACPDU most recently
15          received on this Aggregation Port.";
16     reference
17         "7.3.2.1.34 of IEEE Std 802.1AX";
18 }
19 leaf partner-oper-system-priority {
20     type uint16;
21     description
22         "Indicates the operational value of priority associated
23          with the Partners System ID. The value of this
24          attribute can contain the manually configured value
25          carried in partner-admin-system-priority if there is
26          no protocol Partner.";
27     reference
28         "7.3.2.1.7 of IEEE Std 802.1AX";
29 }
30 leaf partner-oper-system {
31     type ieee:mac-address;
32     description
33         "Represents the MAC address portion of the Aggregation
34          Port's current protocol partner's System Identifier.
35          A value of zero indicates that there is no known
36          protocol Partner.
37          The value of this attribute can contain the manually
38          configured value carried in partner-admin-system if
39          there is no protocol Partner.";
40     reference
41         "7.3.2.1.9 of IEEE Std 802.1AX";
42 }
43 leaf partner-oper-key {
44     type uint16;
45     description
46         "The current operational value of the Key for the
47          protocol Partner. The value of this attribute can
48          contain the manually configured value carried in
49          partner-admin-key if there is no protocol Partner.";
50     reference
51         "7.3.2.1.11 of IEEE Std 802.1AX";
52 }
53 leaf partner-oper-port {
54     type uint16;
55     description
56         "The operational port number assigned by the
57          Aggregation Port's protocol Partner. The value of this
58          attribute can contain the administratively configured
59          value carried in partner-admin-port if there is no
60          protocol Partner.";
61     reference
62         "7.3.2.1.17 of IEEE Std 802.1AX";
63 }
64 leaf partner-oper-port-priority {
65     type uint16;
66     description
67         "The operational priority value assigned by the
68          Aggregation Port's protocol Partner. The value of this
69          attribute can contain the administratively configured
70          value carried in partner-admin-port-priority if there
71          is no protocol Partner.";
72     reference
```

```
1      "7.3.2.1.19 of IEEE Std 802.1AX";
2  }
3  leaf partner-oper-state {
4      type ax:lacp-state;
5      description
6          "The operational value of the partner's LACP state
7           derived from received LACPDUs or, when Defaulted is
8           true, from the partner-admin-state.";
9      reference
10         "7.3.2.1.23, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
11 }
12 leaf aggregate-or-individual {
13     type boolean;
14     description
15         "When true indicates the Aggregation Port can join a
16          LAG consisting of multiple Aggregation Ports.
17          When false, indicates that the Aggregation Port can
18          only operate as an Solitary link because the
19          Aggregation bit is false in either
20          actor-oper-port-state or partner-oper-port-state.";
21     reference
22         "7.3.2.1.24 of IEEE Std 802.1AX";
23 }
24 leaf wtr-waiting {
25     type boolean;
26     description
27         "Indicates the Aggregation Port is inhibited from
28          becoming active for an interval (determined by
29          wtr-time) after becoming operational or while
30          non-revertive operation is being enforced by the
31          Selection Logic.";
32     reference
33         "7.3.2.32 of IEEE Std 802.1AX";
34 }
35 }
36 container cscd {
37     if-feature "ax:cscd";
38     description
39         "Aggregation port parameters for support of CSCD.";
40     leaf admin-link-number {
41         type uint16;
42         description
43             "The Link Number value for the Aggregation Port,
44              that is unique among all Aggregation Ports in the same
45              key group. This leaf provides the ability to
46              administratively override the initial value provided
47              by the system. A value of 0 is allowed, but
48              results in no frames being distributed to this
49              Aggregation Port.
50              The default value is the initial value of actor-port-
51              number provided by the system.";
52         reference
53             "7.3.2.1.27, 6.6.3.2 of IEEE Std 802.1AX";
54     }
55     leaf partner-link-number {
56         type uint16;
57         config false;
58         description
59             "The last received value of the Partner Link Number,
60              or zero if the Aggregation Port is using default
61              values for the Partner or the Partner LACP Version
62              is 1.";
63         reference
64             "7.3.2.1.29 of IEEE Std 802.1AX";
65     }
66     leaf link-number {
67         type uint16;
68         config false;
69         description
70             "The operational link number for this Aggregation Port.
71              The value is either the same as the admin-link-number,
72              or the corresponding value of the LACP partner.";
```



```
1         reference
2         "7.3.2.1.28 of IEEE Std 802.1AX";
3     }
4 }
5 }
6 }
7
8 augment "/if:interfaces/if:interface/if:statistics" {
9     when '../dotlax:aggport' {
10         description
11         "Applies to aggregation ports.";
12     }
13     description
14     "Augment interface statistics with aggport statistics.";
15     container aggport-stats {
16         config false;
17         description
18         "Contains stats associated with the Aggregation Port.";
19         leaf lacp-pdu-rx {
20             type yang:counter64;
21             description
22             "The number of valid LACPDUs received on this
23             Aggregation Port.";
24             reference
25             "7.3.3.1.2 of IEEE Std 802.1AX";
26         }
27         leaf marker-pdu-rx {
28             type yang:counter64;
29             description
30             "The number of valid Marker PDUs received on this
31             Aggregation Port.";
32             reference
33             "7.3.3.1.3 of IEEE Std 802.1AX";
34         }
35         leaf marker-response-pdu-rx {
36             type yang:counter64;
37             description
38             "The number of valid Marker Response PDUs received on
39             this Aggregation Port.";
40             reference
41             "7.3.3.1.4 of IEEE Std 802.1AX";
42         }
43         leaf unknown-rx {
44             type yang:counter64;
45             description
46             "The number of frames received that either:
47             a) Carry the Slow Protocols Ethernet Type value,
48             but contain an unknown PDU, or
49             b) Are addressed to the Slow Protocols group MAC
50             Address, but do not carry the Slow Protocols
51             Ethernet Type.";
52             reference
53             "7.3.3.1.5 of IEEE Std 802.1AX";
54         }
55         leaf illegal-rx {
56             type yang:counter64;
57             description
58             "The number of frames received that carry the Slow
59             Protocols Ethernet Type value, but contain a badly
60             formed PDU or an illegal value of Protocol Subtype.";
61             reference
62             "7.3.3.1.6 of IEEE Std 802.1AX";
63         }
64         leaf lacp-pdu-tx {
65             type yang:counter64;
66             description
67             "The number of LACPDUs transmitted on this
68             Aggregation Port.";
69             reference
70             "7.3.3.1.7 of IEEE Std 802.1AX";
71         }
72         leaf marker-pdu-tx {
```

```
1     type yang:counter64;
2     description
3         "The number of Marker PDUs transmitted on this
4         Aggregation Port.";
5     reference
6         "7.3.3.1.8 of IEEE Std 802.1AX";
7 }
8 leaf marker-response-pdu-tx {
9     type yang:counter64;
10    description
11        "The number of Marker Response PDUs transmitted on
12        this Aggregation Port.";
13    reference
14        "7.3.3.1.9 of IEEE Std 802.1AX";
15 }
16 }
17 }
18
19 container linkagg {
20     description
21         "Link Aaggregation System specific configuration nodes.";
22     list agg-system {
23         key "name";
24         description
25             "List of aggregation systems.";
26         leaf name {
27             type string;
28             description
29                 "Name for the aggregation system.";
30         }
31         leaf actor-system {
32             type ieee:mac-address;
33             description
34                 "The part of the System Identifier that is a globally
35                 unique MAC address. This leaf provides the ability to
36                 administratively override the initial value provided
37                 by the system.";
38             reference
39                 "7.3.1.1.4, 7.3.2.1.3 of IEEE Std 802.1AX";
40         }
41         leaf actor-system-priority {
42             type uint16;
43             default "0x8000";
44             description
45                 "The part of the System Identifier that is the
46                 priority of the system.";
47             reference
48                 "7.3.1.1.5, 7.3.2.1.2 of IEEE Std 802.1AX";
49         }
50     }
51     list key-group {
52         key "name";
53         unique "actor-admin-key";
54         description
55             "List of key groups. A key group is the set of aggregators
56             and aggregation ports that share the same system priority,
57             system identifier, and aggregation key, and therefore can
58             potentially form a Link Aggregation Group. Each entry in
59             the key group list contains the parameters common to all
60             aggregation ports and/or aggregators in the key group.";
61         reference
62             "6.3.5, 6.4.12 of IEEE Std 802.1AX";
63         leaf name {
64             type string;
65             description
66                 "Name for the key group.";
67         }
68         leaf actor-admin-key {
69             type uint16 {
70                 range "1..65535";
71             }
72             mandatory true;
```

```
1      description
2          "The administrative value of the key used by the
3      Aggregators and Aggregation Ports in this key-group.";
4      reference
5          "7.3.1.1.7, 7.3.2.1.4 of IEEE Std 802.1AX";
6  }
7  leaf agg-system-name {
8      type agg-system-ref;
9      mandatory true;
10     description
11         "Specifies the aggregation system for this key
12     group.";
13 }
14 leaf actor-protocol-da {
15     type ax:bridge-reserved-addresses;
16     must ". = 'nearest-customer-bridge' or
17         . = 'slow-protocols-multicast' or
18         . = 'nearest-non-tpmr-bridge'" {
19         error-message "Invalid protocol address";
20     }
21     default "slow-protocols-multicast";
22     description
23         "A 6-octet read-write MAC Address value specifying the DA
24         to be used when sending Link Aggregation Control and
25         Marker PDUs. Valid addresses are the
26         nearest-customer-bridge, slow-protocols-multicast,
27         and nearest-non-tpmr-bridge group addresses. The
28         default value is the slow-protocols-multicast address.";
29     reference
30         "7.3.2.2.1, 6.2.10.2 of IEEE Std 802.1AX";
31 }
32 leaf collector-max-delay {
33     type uint16;
34     description
35         "Specifies the maximum delay, in tens of microseconds,
36         between receiving a frame from an Aggregator Port, and
37         either delivering the frame to the Aggregator Client or
38         discarding the frame. A value of zero means the delay
39         is less than the minimum increment (< 10us).
40         This leaf provides the ability to administratively
41         override the initial value provided by the system.";
42     reference
43         "7.3.1.1.32, 6.2.3.1.1, B.3 of IEEE Std 802.1AX";
44 }
45 leaf actor-admin-state {
46     type ax:lacp-state {
47         bit lacp-activity {
48             position 1;
49             description
50                 "Setting the LACP_Activity state to '0' means that the
51                 transmission of LACPDUs is controlled by the value of
52                 partner-oper-state.LACP_Activity.";
53         }
54         bit lacp-timeout {
55             position 2;
56             description
57                 "Setting the LACP_Timeout to '0' means that actor uses
58                 the Long_Timeout value, allowing the partner to transmit
59                 LACPDUs at the Slow_Periodic_Time.";
60         }
61         bit aggregation {
62             position 3;
63             description
64                 "Setting the Aggregation state to '0' prevents this
65                 port from being aggregated with any other ports.";
66         }
67     }
68     default "lacp-activity lacp-timeout aggregation";
69     description
70         "Provides administrative control over the values of the
71         LACP_Activity, LACP_Timeout, and Aggregation state.";
72     reference
```

```
1      "7.3.2.1.20, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
2  }
3  leaf partner-admin-system {
4      type ieee:mac-address;
5      default "00-00-00-00-00-00";
6      description
7          "The administrative value of the MAC address portion of
8           the Partner's System Identifier.
9           The assigned value is used, along with the value of
10          port-partner-admin-system, partner-admin-key,
11          partner-admin-port, and partner-admin-port-priority,
12          to achieve administratively configured Link
13          Aggregation Groups with a partner that does not run
14          LACP.";
15      reference
16          "7.3.2.1.8 of IEEE Std 802.1AX";
17  }
18  leaf partner-admin-system-priority {
19      type uint16;
20      default "0";
21      description
22          "The administrative value of priority portion of the
23           the Partner's System Identifier. The assigned
24           value is used, along with the value of
25           port-partner-admin-system, partner-admin-key,
26           partner-admin-port, and partner-admin-port-priority,
27           to achieve administratively configured Link
28           Aggregation Groups with a partner that does not run
29           LACP.";
30      reference
31          "7.3.2.1.6 of IEEE Std 802.1AX";
32  }
33  leaf port-algorithm {
34      type identityref {
35          base ax:distribution-algorithm;
36      }
37      default "ax:unspecified";
38      description
39          "Identifies the algorithm used by the Aggregator to
40           assign frames to a Port Conversation ID. Default is
41           the value for an unspecified distribution algorithm.
42           When the identity description specifies a 4 octet value,
43           this value will be used for cscd purposes, and included
44           in LACPDUs. Otherwise this leaf is used purely for
45           local selection of a distribution algorithm, and the
46           unspecified distribution algorithm is used for cscd
47           purposes.";
48      reference
49          "7.3.1.1.33, Table 8-1 of IEEE Std 802.1AX";
50  }
51  leaf-list lags {
52      type if:interface-ref;
53      config false;
54      description
55          "A list of the if:name of aggregators assigned to this
56           key group.";
57      reference
58          "linkagg:key-group";
59  }
60  leaf-list aggports {
61      type if:interface-ref;
62      config false;
63      description
64          "A list of the if:name of aggregation ports assigned to
65           this key group.";
66      reference
67          "linkagg:key-group";
68  }
69  container cscd {
70      if-feature "ax:cscd";
71      description
72          "Contains CSCD parameters that need to be consistent for
```

```
1      all aggregation ports and aggregators in the key group.";
2  container admin-conv-service-map {
3      if-feature "ax:sid-map";
4      description
5          "Data structure to map service identifiers to
6          conversation identifiers. Each entry consists of a
7          Conversation ID (CID) and a list of zero or more
8          Service Identifiers (SIDs) that map to it. An empty
9          list of SIDs means there are no SIDs that map to this
10         CID, and results in the same behavior as not having an
11         entry for this CID.";
12     reference
13         "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
14     uses ax:service-map;
15 }
16 leaf admin-conv-service-digest {
17     type binary;
18     config false;
19     description
20         "The MD5 Digest of the admin-conv-service-map. The
21         value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64)
22         when the distribution algorithm specified by
23         agg-port-algorithm does not use the
24         admin-conv-service-map.";
25     reference
26         "7.3.1.1.39, 6.6.3.1 of IEEE Std 802.1AX";
27 }
28 container admin-conv-link-map {
29     description
30         "Data structure to map a Conversation Identifier
31         (CID) to a Link Number. Each entry consists of a CID
32         and a list of link numbers that can potentially be
33         selected for that CID. An empty list of link-numbers
34         means that no links are selected for the CID.";
35     reference
36         "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
37     uses ax:link-map;
38 }
39 leaf admin-conv-link-digest {
40     type binary;
41     config false;
42     description
43         "The MD5 Digest of the admin-conv-link-map. The value
44         value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64)
45         when the distribution algorithm specified by
46         agg-port-algorithm does not use the
47         admin-conv-link-map.";
48     reference
49         "7.3.1.1.38, 6.6.3.1 of IEEE Std 802.1AX";
50 }
51 leaf admin-discard-wrong-conv {
52     if-feature "ax:dwc";
53     type enumeration {
54         enum force-true {
55             value 1;
56             description
57                 "Indicates that an Aggregator discards a
58                 frame that is collected from an Aggregation Port
59                 that is different from the Aggregation Port to
60                 which the Aggregator would distribute a frame
61                 with the same Port Conversation ID.";
62         }
63         enum force-false {
64             value 2;
65             description
66                 "Indicates that an Aggregator does not discard
67                 a frame that is collected from an Aggregation Port
68                 that is different from the Aggregation Port to
69                 which the Aggregator would distribute a frame with
70                 the same Port Conversation ID. This is the behavior
71                 of the Aggregator when DWC is not supported";
72         }
73     }
74 }
```

```

1      enum auto {
2          value 3;
3          description
4              "Indicates that the Aggregator behaves as
5              if the value was force-true only when the actor
6              and partner agree on the algorithms (other than
7              unspecified) and mapping tables used to map frames
8              to Aggregation Ports, and behaves as if the value
9              was force-false otherwise.";
10     }
11 }
12 default "force-false";
13 description
14     "Indicates whether an Aggregator discards a
15     frame that is collected from an Aggregation Port
16     that is different from the Aggregation Port to which
17     the Aggregator would distribute a frame with the
18     same Port Conversation ID.";
19 reference
20     "7.3.1.1.35, 6.6 of IEEE Std 802.1AX";
21 }
22 }
23 }
24 }
25 }
26
27

```

28 10.7.3 The *ieee802-dot1ax-drni* YANG module

```

29 module ieee802-dot1ax-drni {
30     yang-version 1.1;
31     namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-drni";
32     prefix dot1ax-drni;
33
34     import ieee802-dot1ax-types {
35         prefix ax;
36     }
37     import ieee802-dot1ax-linkagg {
38         prefix dot1ax;
39     }
40     import ieee802-types {
41         prefix ieee;
42     }
43     import ietf-yang-types {
44         prefix yang;
45     }
46     import ietf-interfaces {
47         prefix if;
48     }
49
50     organization
51         "IEEE 802.1 Working Group";
52     contact
53         "WG-URL: http://www.ieee802.org/1/
54         WG-Email: stds-802-1-1@ieee.org
55
56         Contact: IEEE 802.1 Working Group Chair
57         Postal: C/O IEEE 802.1 Working Group
58                 IEEE Standards Association
59                 45 Hoes Lane
60                 Piscataway, NJ 08854
61                 USA
62         E-mail: stds-802-1-chairs@ieee.org";
63     description
64         "This YANG module describes the configuration model for a
65         Distributed Resilient Network Interface (DRNI) as specified
66         in 802.1AX.
67
68         Copyright (C) IEEE (2025).
69

```

```
1   This version of this YANG module is part of IEEE Std 802.1AX;
2   see the standard itself for full legal notices.";
3
4   revision 2025-03-11 {
5     description
6       "Published as part of IEEE Std 802.1AXdz-202y.
7        The following reference statement identifies each referenced IEEE
8        Standard as updated by applicable amendments.";
9     reference
10      "IEEE Std 802.1AX-2020 Link Aggregation:
11       IEEE Std 802.1AXdz-202y.";
12   }
13
14   typedef irp-state {
15     type bits {
16       bit reserved-1 {
17         position 1;
18         description
19           "Reserved for future use. It is set to 0 on
20            transmit and ignored on receipt.";
21       }
22       bit reserved-2 {
23         position 2;
24         description
25           "Reserved for future use. It is set to 0 on
26            transmit and ignored on receipt.";
27       }
28       bit short-timeout {
29         position 3;
30         description
31           "The Short_Timeout flag indicates the Timeout control value
32            in use by the DRCP Receive machine on this IRP. Short Timeout
33            is encoded as a 1; Long Timeout is encoded as a 0.";
34       }
35       bit synchronization {
36         position 4;
37         description
38           "When the Sync flag is TRUE (1), the DRCP Receive machine has
39            determined the Neighbor DRNI System has a compatible
40            configuration for forming a DRNI.";
41       }
42       bit irc-data {
43         position 5;
44         description
45           "When the IRC Data flag is TRUE (1), the transfer of Up
46            and Down frames is permitted on the IRC.";
47       }
48       bit drni {
49         position 6;
50         description
51           "The DRNI flag is TRUE (1) when this DRNI System is paired
52            with another DRNI System and FALSE (0) otherwise.";
53       }
54       bit defaulted {
55         position 7;
56         description
57           "When the Defaulted flag is TRUE (1), the DRCP Receive machine
58            is using default operational Neighbor information.
59            When FALSE (0), the operational Neighbor information
60            in use has been received in a DRCPDU.";
61       }
62       bit expired {
63         position 8;
64         description
65           "When the Expired flag is TRUE (1), the DRCP Receive machine
66            is in the EXPIRED state.";
67       }
68     }
69     description
70       "IRP state values as transmitted in DRCPDUs.
71        Administrative control over the values of Short_Timeout
72        and IRC_Data is allowed. The remaining bits are read-only.";
```

```

1  reference
2  "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
3  }
4
5  grouping drni-mask {
6    description
7    "Specifies the contents of a bit mask indexed by CID.";
8    leaf-list cid-str {
9      type string {
10       pattern '(0|[1-9][0-9]{0,3})'
11       + '((-0|[1-9][0-9]{0,3})'
12       + '(%0|[1-9][0-9]{0,3}))?)?'
13       + '(',
14       + '(0|[1-9][0-9]{0,3})'
15       + '((-0|[1-9][0-9]{0,3})'
16       + '(%0|[1-9][0-9]{0,3}))?)?'
17       + ')*';
18     }
19     default "0-4095";
20     description
21     "A comma separated list of single CID values (x),
22     CID ranges (x-y), or ranges with an increment (x-y%z).
23     For example, the string '0,10-13,100-110%4' sets the
24     bits in the mask corresponding to CIDs
25     0,10,11,12,13,100,104,108.
26     Some special cases are:
27     'x-y%z' where x+z>y is the same as 'x'.
28     'x-y%0' is the same as 'x'.
29     'x-y%1' is the same as 'x-y'.
30     'x-y%2' sets the bit for all even CIDs in the range
31     when x is even, and for all odd CIDs when x is odd.
32     Although the pattern allows values of x, y, and z in the
33     range 0-9999, resulting CID values greater than 4095
34     have no effect on the mask.
35     Specified as a list of strings, so that long strings
36     can be broken into several shorter strings.
37     The default sets all bits in the mask.";
38   }
39   leaf invert-mask {
40     type boolean;
41     default "false";
42     description
43     "When true the specified mask is inverted:
44     each zero replaced with a one,
45     and each one replaced with a zero.";
46   }
47 }
48
49 augment "/if:interfaces/if:interface/dot1ax:lag" {
50   description
51   "Augmentation parameters only for Aggregators with
52   DRNI enabled.";
53   container drni {
54     presence "When present, this Aggregator is enabled for DRNI";
55     description
56     "Aggregator parameters to support a Distributed
57     Resilient Network Interface";
58     leaf irp-name {
59       type if:interface-ref;
60       mandatory true;
61       description
62       "Interface Name (if:name) of the Port supporting the
63       Intra Relay Port (IRP) of this DRNI Gateway.";
64       reference
65       "7.4.1.1.23 of IEEE Std 802.1AX";
66     }
67     leaf drni-aggregator-key {
68       type uint16 {
69         range "1..65535";
70       }
71       description
72       "The Aggregator Key value to be used by the Aggregator

```



```
1      supporting this DRNI Gateway (and the Aggregation Ports
2      assigned to this DRNI Gateway) when paired with a
3      neighbor DRNI System via the IRC. The default value is
4      the Aggregator's Actor_Admin_Aggregator_Key value.";
5      reference
6      "7.4.1.1.15 of IEEE Std 802.1AX";
7  }
8  leaf drni-aggregator-system {
9      type ieee:mac-address;
10     default "00-00-00-00-00-00";
11     description
12         "The Aggregator System value to be used by the
13         Aggregator supporting this DRNI Gateway (and the
14         Aggregation Ports assigned to this DRNI Gateway)
15         when paired with a neighbor DRNI System via the
16         Intra-Relay Connection (IRC).";
17     reference
18         "7.4.1.1.13 of IEEE Std 802.1AX";
19 }
20 leaf drni-aggregator-system-priority {
21     type uint16;
22     default "0";
23     description
24         "The Aggregator System Priority value to be used by the
25         Aggregator supporting this DRNI Gateway (and the
26         Aggregation Ports assigned to this DRNI Gateway) when
27         paired with a neighbor DRNI System via the IRC.";
28     reference
29         "7.4.1.1.14 of IEEE Std 802.1AX";
30 }
31 leaf drcp-protocol-da {
32     type ax:bridge-reserved-addresses;
33     must "." = 'nearest-customer-bridge' or
34         . = 'nearest-bridge' or
35         . = 'nearest-non-tpmr-bridge' {
36         error-message "Invalid protocol address";
37     }
38     default "nearest-non-tpmr-bridge";
39     description
40         "A 6-octet read-write MAC Address value specifying the
41         Destination Address for Distributed Relay Control PDUs
42         transmitted on the Intra-Relay Port. Valid addresses are
43         the nearest-customer-bridge, nearest-bridge, and
44         nearest-non-tpmr-bridge group addresses. The default
45         value is the nearest-non-tpmr-bridge group
46         address.";
47     reference
48         "7.4.1.1.12, 9.6.1.1 of IEEE Std 802.1AX";
49 }
50 leaf home-admin-irp-state {
51     type irp-state {
52         bit short-timeout {
53             position 3;
54             description
55                 "The Short_Timeout flag indicates the Timeout control value
56                 in use by the DRCP Receive machine on this IRP. Short Timeout
57                 is encoded as a 1; Long Timeout is encoded as a 0.";
58         }
59         bit irc-data {
60             position 5;
61             description
62                 "When the IRC_Data flag is TRUE (1), the transfer of Up
63                 and Down frames is permitted on the IRC.";
64         }
65     }
66     default "short-timeout irc-data";
67     description
68         "Provides administrative control over the values of the
69         Short_Timeout and IRC-Data state.";
70     reference
71         "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
72 }
```

```
1 leaf home-oper-irp-state {
2   type irp-state;
3   config false;
4   description
5     "A string of 8 bits, corresponding to the current
6     operational value of IRP_State as transmitted in
7     DRCPDUs.";
8   reference
9     "7.4.1.1.25, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
10 }
11 leaf home-cscd-gateway-control {
12   type boolean;
13   default "true";
14   description
15     "When TRUE, allows the DRNI Gateway Port selection to
16     be based on the CSCD parameters that control the
17     Aggregator Port selection.";
18   reference
19     "7.4.1.1.16 of IEEE Std 802.1AX";
20 }
21 leaf home-dr-client-gateway-control {
22   type boolean;
23   default "true";
24   description
25     "When TRUE, allows the Distributed Relay Client to
26     determine whether to forward frames through the DRNI
27     Gateway Port.";
28   reference
29     "7.4.1.1.17 of IEEE Std 802.1AX";
30 }
31 leaf home-gateway-algorithm {
32   type identityref {
33     base ax:distribution-algorithm;
34   }
35   default "ax:unspecified";
36   description
37     "Identifies the algorithm used by the DRNI Gateway to
38     assign frames to a Gateway Conversation ID. 8.2 provides
39     the IEEE 802.1 OUI (00-80-C2) Gateway Algorithm
40     encodings. Default is the value for an unspecified
41     distribution algorithm.";
42   reference
43     "7.4.1.1.6 of IEEE Std 802.1AX";
44 }
45 container home-gateway-conv-service-map {
46   if-feature "ax:sid-map";
47   description
48     "Data structure to map service identifiers to
49     conversation identifiers. Each entry consists of a
50     Conversation ID (CID) and a list of zero or more Service
51     Identifiers (SIDs) that map to it. Frames with Service
52     IDs not contained in the map are not mapped to any
53     Gateway Conversation ID and are discarded.";
54   reference
55     "7.4.1.1.20, 6.6.3.1 of IEEE Std 802.1AX";
56   uses ax:service-map;
57 }
58 leaf home-gateway-conv-service-digest {
59   type binary;
60   config false;
61   description
62     "The MD5 Digest of the home-gateway-conv-service-map. The
63     value is NULL (AAAAAAAAAAAAAAAAAAAAA== in base64) when
64     the distribution algorithm specified
65     by agg-port-algorithm does not use the
66     home-gateway-conv-service-map.";
67   reference
68     "7.4.1.1.21 of IEEE Std 802.1AX";
69 }
70 container gateway-enable-mask {
71   description
72     "A vector of Boolean values, with one value for each
```

```
1         possible Gateway Conversation ID. A 1 indicates that
2         frames associated with that Gateway Conversation ID
3         are allowed to pass through this Gateway Port, and a
4         0 indicates that such frames are not allowed to pass.
5         Default value is all bits set to 1.";
6     reference
7         "7.4.1.1.18, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
8     uses drni-mask;
9 }
10 container gateway-preference-mask {
11     description
12         "A vector of Boolean values, with one value for each
13         possible Gateway Conversation ID. A 1 indicates that
14         this Gateway Port is the preferred Gateway when both
15         DRNI Gateways have the Gateway Conversation ID enabled
16         in the gateway-available-mask, and a 0 indicates that
17         it is not preferred.
18         Default value is all bits set to 1.";
19     reference
20         "7.4.1.1.19, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
21     uses drni-mask;
22 }
23 container gateway-available-mask {
24     config false;
25     description
26         "A vector of Boolean values, with one value for each
27         possible Gateway Conversation ID. A 1 indicates that
28         this Gateway Port is eligible to be selected to pass
29         that Gateway Conversation ID, and a 0 indicates that
30         it is not eligible.";
31     reference
32         "7.4.1.1.22, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
33     uses drni-mask;
34 }
35 container neighbor {
36     config false;
37     description
38         "Operational values for the DRNI neighbor obtained
39         from DRCPDUs.";
40     leaf oper-irp-state {
41         type irp-state;
42         description
43             "A string of 8 bits, corresponding to the current
44             operational value of IRP_State as transmitted in
45             DRCPDUs.";
46         reference
47             "7.4.1.1.26, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
48     }
49     leaf system {
50         type ieee:mac-address;
51         description
52             "The MAC Address portion of the System Identifier of
53             the Neighbor DRNI System (connected via the
54             Intra-Relay Port). ";
55         reference
56             "7.4.1.1.29 of IEEE Std 802.1AX";
57     }
58     leaf system-priority {
59         type uint16;
60         description
61             "The priority portion of the System Identifier of the
62             Neighbor DRNI System (connected via the Intra-Relay
63             Port).";
64         reference
65             "7.4.1.1.30 of IEEE Std 802.1AX";
66     }
67     leaf drni-key {
68         type uint16;
69         description
70             "The DRNI key value received from the Neighbor DRNI
71             System (connected via the IntraRelay Port).";
72         reference
```

```
1      "7.4.1.1.31 of IEEE Std 802.1AX";
2  }
3  leaf aggregator-algorithm {
4      type identityref {
5          base ax:distribution-algorithm;
6      }
7      description
8          "The Port algorithm used by the Neighbor Aggregator to
9          assign frames to Port Conversation IDs.";
10     reference
11         "7.4.1.1.33 of IEEE Std 802.1AX";
12 }
13 leaf aggregator-conv-service-digest {
14     type binary;
15     description
16         "The MD5 Digest of the Neighbor Aggregator's
17         Admin_Conv_Service_Map. Obtained from the Home
18         Aggregator State TLV last received from the Neighbor
19         DRNI System.";
20     reference
21         "7.4.1.1.34 of IEEE Std 802.1AX";
22 }
23 leaf aggregator-conv-link-digest {
24     type binary;
25     description
26         "The MD5 Digest of the Neighbor Aggregator's
27         Admin_Conv_Link_Map. Obtained from the Home Aggregator
28         State TLV (9.6.2.4) last received from the Neighbor
29         DRNI System.";
30     reference
31         "7.4.1.1.35 of IEEE Std 802.1AX";
32 }
33 leaf partner-system-priority {
34     type uint16;
35     description
36         "The priority portion of the System Identifier of the
37         Neighbor Aggregator's Partner.";
38     reference
39         "7.4.1.1.36 of IEEE Std 802.1AX";
40 }
41 leaf partner-system {
42     type ieee:mac-address;
43     description
44         "The MAC Address portion of the System Identifier of
45         the Neighbor Aggregator's Partner.";
46     reference
47         "7.4.1.1.37 of IEEE Std 802.1AX";
48 }
49 leaf partner-aggregator-key {
50     type uint16;
51     description
52         "The operational key value of the Neighbor
53         Aggregator's Partner.";
54     reference
55         "7.4.1.1.38 of IEEE Std 802.1AX";
56 }
57 leaf cscd-state {
58     type bits {
59         bit reserved-1 {
60             position 0;
61             description
62                 "Bit 1 is reserved for future use. It is set to 0
63                 and ignored on receipt.";
64         }
65         bit reserved-2 {
66             position 1;
67             description
68                 "Bit 2 is reserved for future use. It is set to 0
69                 and ignored on receipt.";
70         }
71         bit reserved-3 {
72             position 2;
```

```

1      description
2          "Bit 3 is reserved for future use. It is set to 0
3          and ignored on receipt.";
4      }
5      bit cscd_gateway_control {
6          position 3;
7          description
8              "CSCD_Gateway_Control is encoded in bit 4. When
9              this flag is TRUE, the DRNI Gateway is configured
10             to minimize forwarding data frames on the IRC by
11             selecting the DRNI Gateway and Aggregator Ports
12             for forwarding any given Conversation ID to be in
13             the same DRNI System.";
14     }
15     bit discard_wrong_conversation {
16         position 4;
17         description
18             "Discard_Wrong_Conversation is encoded in bit 5.
19             The Aggregator's Discard_Wrong_Conversation
20             value.";
21     }
22     bit differ_conv_link_digests {
23         position 5;
24         description
25             "Differ_Conv_Link_Digests is encoded in bit 6.
26             This flag is TRUE when the Aggregator's
27             Actor_Conv_Link_Digest matches the Aggregator's
28             Partner_Conv_Link_Digest.";
29     }
30     bit differ_conv_service_digests {
31         position 6;
32         description
33             "Differ_Conv_Service_Digests is encoded in bit 7.
34             This flag is TRUE when the Aggregator's
35             Actor_Conv_Service_Digest matches the Aggregator's
36             Partner_Conv_Service_Digest.";
37     }
38     bit differ_port_algorithms {
39         position 7;
40         description
41             "Differ_Port_Algorithms is encoded in bit 8. The
42             Aggregator's differPortAlgorithms flag is TRUE
43             when the Aggregator's Actor_Port_Algorithm matches
44             the Aggregator's Partner_Port_Algorithm.";
45     }
46 }
47 description
48     "8 bits, corresponding to the Aggregator_CSCD_State
49     in the Neighbor_Aggregator_State variable. The first
50     three bits (the least significant bits of CSCD_State)
51     are reserved; the fourth bit corresponds to the
52     Neighbor's value for Home_Admin_CSCD_Gateway_Control;
53     the fifth bit corresponds to the Neighbor Aggregator's
54     operational value for Discard_Wrong_Conversation; and
55     the sixth, seventh, and eighth bits correspond to the
56     Neighbor Aggregator's operational value for
57     differConvLinkDigests, differConvServiceDigests, and
58     differPortAlgorithms, respectively, (the most
59     significant bits of CSCD_State).";
60 reference
61     "7.4.1.1.39 of IEEE Std 802.1AX";
62 }
63 leaf-list active-links {
64     type uint16;
65     description
66         "A list of the operational Link_Numbers of Aggregation
67         Ports that are currently active (i.e., collecting) on
68         the Neighbor's Aggregator. An empty list indicates that
69         there are no Aggregation Ports active. Each integer
70         value in the list carries an aAggPortOperLinkNumber
71         attribute value.";
72     reference

```

```

1      "7.4.1.1.40 of IEEE Std 802.1AX";
2    }
3    leaf gateway-algorithm {
4      type identityref {
5        base ax:distribution-algorithm;
6      }
7      description
8        "The gateway algorithm used by the Neighbor DRNI
9        Gateway to assign frames to Gateway Conversation IDs.";
10     reference
11       "7.4.1.1.41 of IEEE Std 802.1AX";
12   }
13   leaf gateway-conv-service-digest {
14     type binary;
15     description
16       "The MD5 Digest of the Neighbor DRNI Gateway's
17       the Home_Admin_Gateway_Conv_Service_Map. Obtained
18       from Gateway_Conv_Service_Digest in the
19       Neighbor_Gateway_State TLV last received from the
20       Neighbor_DRNI System.";
21     reference
22       "7.4.1.1.42 of IEEE Std 802.1AX";
23   }
24   container gateway-available-mask {
25     description
26       "A vector of Boolean values, with one value for each
27       possible Gateway Conversation ID. A 1 indicates that
28       the Neighbor DRNI Gateway Port is eligible to be
29       selected to pass that Gateway Conversation ID, and
30       a 0 indicates that it is not eligible.";
31     reference
32       "7.4.1.1.43 of IEEE Std 802.1AX";
33     uses drni-mask;
34   }
35   container gateway-preference-mask {
36     description
37       "A vector of Boolean values, with one value for each
38       possible Gateway Conversation ID. A 1 indicates that
39       the Neighbor DRNI Gateway Port is the preferred
40       Gateway when both DRNI Gateways have the Gateway
41       Conversation ID enabled in the gateway-available-mask,
42       and a 0 indicates that it is not preferred.";
43     reference
44       "7.4.1.1.44 of IEEE Std 802.1AX";
45     uses drni-mask;
46   }
47 }
48 }
49 }
50
51 augment "/if:interfaces/if:interface/if:statistics"
52 + "/dotlax:lag-stats" {
53   when '../../dotlax:lag/dotlax-drni:drni' {
54     description
55       "Applies to aggregators with DRNI present.";
56   }
57   description
58     "Augment interface statistics with DRNI statistics.";
59   container drni-stats {
60     description
61       "Contains DRNI specific statistics.";
62     leaf drcpdus-rx {
63       type yang:counter64;
64       description
65         "The number of valid DRCPDUs received on this
66         Intra-Relay Port.";
67       reference
68         "7.4.1.1.45 of IEEE Std 802.1AX";
69     }
70     leaf illegal-rx {
71       type yang:counter64;
72       description

```

```
1      "The number of frames received on this Intra-Relay
2      Port that carry the DRCP EtherType value,
3      but contain a badly formed PDU.";
4      reference
5      "7.4.1.1.46, 9.6.1.4 of IEEE Std 802.1AX";
6  }
7  leaf drcpdus-tx {
8      type yang:counter64;
9      description
10     "The number of valid DRCPDUs transmitted on this
11     Intra-Relay Port.";
12     reference
13     "7.4.1.1.47 of IEEE Std 802.1AX";
14 }
15 }
16 }
17 }
18
19
20
```

¹ Annex A

² (normative)

³ Protocol implementation conformance statement (PICS) ⁴ proforma⁷

⁵ Insert the following rows at the end of the table in A.2.1

A.2.1 Major capabilities/options

Item	Feature	Status	References	Support
YANG	Does the implementation support management operations using YANG modules?	O	10.6	Yes [] No []
YANG modules	Is the ieee802-dot1ax-types module supported?	YANG:M	10.7.1	N/A [] Yes []
	Is the ieee802-dot1ax-linkagg module supported?	YANG:M	10.7.2	N/A [] Yes []
	Is the ieee802-dot1ax-drni module supported?	DRNI AND YANG:M	10.7.3	N/A [] Yes []

⁶

⁷ Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

¹ **Annex G**

² (informative)

³ **Bibliography**

⁴ *Insert the following two bibliography entries after [B8] in Annex G:*

⁵

⁶ [B9] IETF RFC 7803, Changing the Registration Policy for the NETCONF Capability URNs Registry,
⁷ February 2016.

⁸ [B10] IETF RFC 8040, RESTCONF Protocol, January 2017.

⁹ [B11] OMG Unified Modeling Language (OMG UML), Version 2.5, March 2015.⁸

⁸OMG documents are available from the Object Management Group (<https://www.omg.org/>).